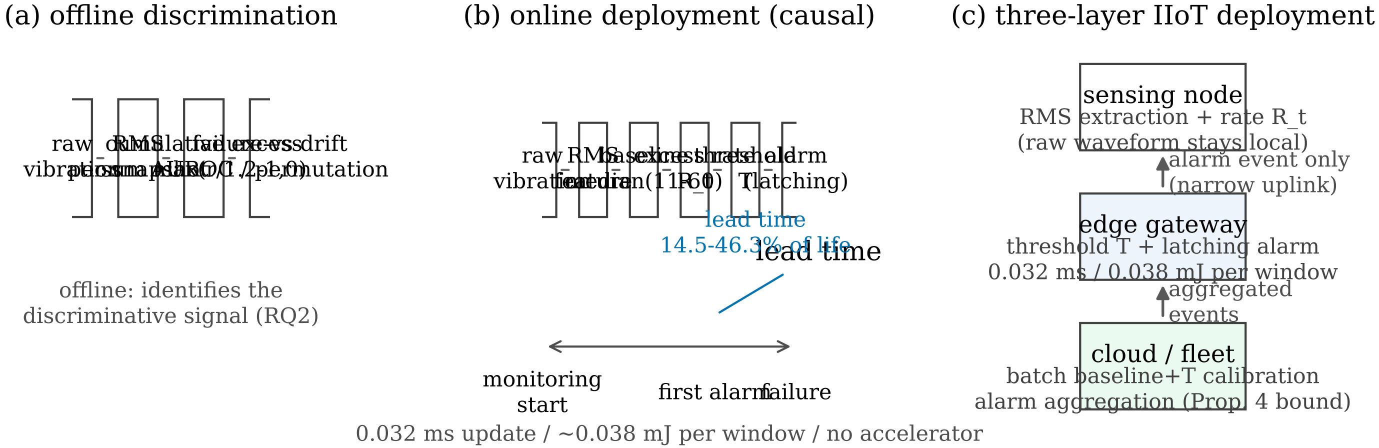
(a) offline discrimination
raw vibration
RMS
permutation
offline: identifies the discriminative signal (RQ2)
(b) online deployment (causal)
raw vibration
RMS feature
baseline median
excess rate R_t
threshold T
alarm (latching)
lead time
14.5-46.3% of life
lead time
monitoring start
first alarm
failure
0.032 ms update / ~0.038 mJ per window / no accelerator
(c) three-layer IIoT deployment
sensing node
RMS extraction + rate R_t
(raw waveform stays local)
alarm event only
(narrow uplink)
edge gateway
threshold T + latching alarm
0.032 ms / 0.038 mJ per window
aggregated events
cloud / fleet
batch baseline+T calibration
alarm aggregation (Prop. 4 bound)